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Multi-modal Understanding of Human Heart (S3.5-COM-CChen2)

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Chapter 1

Introduction

Cardiovascular disease is a significant cause of mortality and morbidity, accounting for a quarter of all deaths in the UK. It is the leading global cause of death, with an estimated 18 million deaths each year (over 30% of all global deaths) [1]. In the current landscape, there have been improvements in medical imaging, such as cardiac magnetic resonance imaging (MRI), computed tomography (CT), as well as non-imaging data like electrocardiogram (ECG), that allows for detailed observation of the cardiovascular system. But, this only captures a partial view of cardiac health.

In practice, clinical assessment requires multiple sources of data to make judgement. Often, clinicians use electrocardiograms and background information, such as demographic and patient medical history, to create an understanding of the patient's cardiovascular system. Despite this fact, a lot of computational approaches to cardiac modelling and analysis operate on individual modalities in isolation [2]. This limits the ability to capture complex relations that are brought by the multiple factors that exist in cardiovascular diseases. Multi-modal machine learning methods have shown potential in clinical settings simply because clinicians must consider multiple data sources into decision-making processes, enabling joint reasoning over imaging and non-imaging data [3]. Generative modelling enables for learning of structured representations of anatomical physiological functions, which can be used for reconstruction, simulation and uncertainty quantification.

Significant challenges still remain before these methods can be considered reliable in clinical settings. Medical data can be noisy, incomplete and inconsistent, causing data-driven models to possibly fail to acknowledge the physiological constraints, which undermines clinical trust and its interpretation. These challenges require an AI systems that are both data efficient and clinically informed, integrating multi-modal information and including current knowledge into model design.

This research proposal aims to look at existing work on cardiac imaging and modelling, use of multi-modal learning in healthcare and generative approaches for reconstruction of

cardiac shape and function.

Chapter 2

Literature Survey

2.1 Background

Current clinical settings use various data sources to make informed decisions on patient condition. These vary from imaging data, which include the use MRIs and CT scans, as well as non-imaging data such as electrocardiogram (ECG). However, they are used as single modals to make these decisions. The use of multi-modal data makes the decision process faster and more accurate, hence why it is the scope of the research project.

2.1.1 Magnetic Resonance Imaging (MRI)

MRIs function by using non-ionising electromagnetic radiation and uses radio frequency radiation in controlled magnetic fields to produce high quality images of the body in any plane [4]. In cardiology, MRIs helps visualise the structure by delineating the size and shape of cardiac chambers and vascular structured in great detail. It also accurately measures ventricular volumes, ejection fraction and wall motion and thickness. MRIs are great at tissue characterisation, where it can differentiate tissue types and disease states. However, there are limitations of current cardiac MRI practice, including variability, incomplete data, and challenges in interpretation, which creates a need for generative, multi-modal, and clinically informed AI methods to produce robust and personalised heart models [5].

2.1.2 Computed Tomography (CT)

Cardiac computed tomography combines multi-detector X-ray acquisition with ECG gating to minimise cardiac motion. When the coronary artery has been iodinated, for contrast visualisation, it can be visualised with high resolution, creating a three-dimensional reconstruction of the coronary tree. This approach gives great detail of the anatomy. When comparing cardiac CT and cardiac MRI, their strengths lie in different domains. Cardiac CT is strongest in coronary anatomy and plaque, whereas cardiac MRI is very good for ventricular function and tissue characterisation. By integrating both modalities gives birth to an idea of multi-imaging modalities which can be represented in cardiac anatomy and function [6].

2.1.3 Electrocardiogram (ECG)

The electrocardiogram is a non-imaging diagnostic recording of the electrical activity of the heart taken from electrodes on the surface of the body. The impulses originate from the sinoatrial node (SAN) and propagate through the atria, which travels through the His–Purkinje system to activate the ventricles. Depolarization and repolarization of cardiac cells create tiny electrical currents. The voltage create a time-series signal, known as the P-QRS-T curve, which reflects the timing and direction of cardiac electrical events. The wave shapes show physiological information about conduction, rhythm and cardiac structure. Clinicians are able to infer abnormalities like arrhythmias and myocardial infarction from the disturbances in the signal patterns. Unlike imaging modalities that show structure or motion, the ECG directly measures the electrical dynamics of the heart. This provides information that may be invisible in images alone. ECG is used as complementary functional information and are often the first indicator of cardiac dysfunction. Hence, why it would be a valuable component of a multi-modal system [7].

2.1.4 Summary

Both imaging and non-imaging data provide great insight into the conditions of the heart. However, the issue arises when single modality becomes inconsistent and incomplete, where noise and uncertainty exists due to the data, hence the lack of robust methods that can learn from imperfect clinical data. Thus, the solution to use multi-modal data to create robust data-efficient methods to build detailed three-dimensional models of the heart.

2.2 Multi-modal learning in AI

In order to incorporate multi-modal learning for AI learning strategies, a strategy of fusing the data together must be integrated in order to feed the model to personally classify heart function to support precise treatment plans.

2.2.1 Fusion Strategies

Various fusion strategies exist and each have their different purposes depending on the data being used, efficiency reasons and model being used.

Early Fusion

In early fusion, the features are extracted from the input data and are then combined to be used as an input to a single analysis unit to perform the analysis. This fusion level is useful for the identification of correlation between multiple features belonging to different modalities at an early stage, helping in better task accomplishment. One phase is also needed for learning the combined feature vector. However, it is hard to represent time synchronisation between multimodal features, as features from different modals could be extracted at different times.

The increase in the number of modalities increases the difficulty to learn the cross correlation between heterogeneous features [8]

Late Fusion

In late fusion, the analysis units provide local decisions that obtained from individual features, which are then combined using a decision fusion unit to make a fused decision vector that is analysed even more to make a final decision about the task at hand. As it is being fused late, the advantage of using late fusion is that the decisions usually have the same representation, making the fusion of decisions easier. Another advantage would be that it allows for the most suitable methods for analysing each single modality. However, issues arise with late fusion in its problem to use feature level correlation among modalities. Also, as different classifiers are used to get local decisions, the learning process can be inefficient [8].

Intermediate Fusion

Early and late fusion both have their advantages and disadvantages. To exploit both, it is possible to use both feature and decision level strategies, when individual features are being analysed by different analysis units, their decisions are fused using a decision fusion unit. All decisions made from previous stages are fused by a decision fusion unit to form a final decision. This strategy obtains the advantages of early and late fusion [8].

Challenges

When applying multimodal learning to reconstructing the cardiac shape, there are challenges to integrating heterogeneous data and fusing information across modalities.

In heterogeneous data, different data structures make it difficult to combine data to a unified representation. Before fusion, normalisation and preprocessing are required for consistency across modalities. The fusion strategy requires careful selection as each fusion strategy has their advantages and disadvantages, and intermediate fusion requires careful architectural design. Implementation to the real world can be limited, where tested systems are on pre-existing public datasets rather than on clinical environments, where noisy and incomplete patient data exists [9].

In a PCI outcome prediction study [10], intermediate fusion was implemented by first learning representations for clinical, biological, and procedural data, which were subsequently fused at the feature level to enable joint risk modelling. This approach allowed the model to capture cross-modal interactions that were inaccessible to early fusion, which suffered from dimensionality and missing-data issues, and to late fusion, which combined modality-specific predictions without shared representation learning. Intermediate fusion achieved superior predictive performance while remaining compatible with real-world clinical data heterogeneity.

2.3 Generative Models for Cardiac Structure & Function

Once the modalities have been fused, the data then can be used to form a model that is designed to learn the underlying patterns and structure of a dataset and generate new data points that could be part of the original dataset, this is referred to as Generative AI. It is a benefit for generative models to create new data that is not limited to the original training dataset. In medical imaging, the use case of generative AI is that it can identify anomalies in medical images, which is useful for diagnosing diseases.

2.3.1 Diffusion Models

One example of generative models that helps with the problem of noisy and incomplete data is diffusion models. Diffusion models alleviate the restriction of class imbalance due to the rare nature of some pathologies by generating diverse realistic-looking images [11]. Chung and Ye [12] conducted a study for a score-based diffusion framework for accelerated MRI reconstruction. This approach outperforms or matches state-of-the-art supervised methods while offering uncertainty quantification and strong generalisation when reconstructing undersampled knee MRI images. The main challenge was integrating the diffusion model with strict MRI data consistency constraints using priors trained only on magnitude images to ensure generalisation across unseen sampling patterns.

2.4 Open Challenges & Research Gaps

In most cardiovascular AI systems, they operate on single modalities as opposed to multiple. If multiple is used, shallow fusion strategies were implemented in using multiple data sources. Multimodal learning reviews consistently show that early and late fusion tend to be mostly used, despite their limitations [10]. When addressing the noisy and incomplete data problem, diffusion models do tend to be used but only to denoise and reconstruction and are rarely used for modelling of the heart. This project addresses these challenges by developing diffusion-based multimodal generative models that capture cardiac structure and enabling personalised assessment of cardiovascular disease across diverse clinical settings.

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